

RESEARCH ARTICLE | APRIL 17 2015

Cluster dynamics models of irradiation damage accumulation in ferritic iron. II. Effects of reaction dimensionality

Aaron A. Kohnert; Brian D. Wirth



J. Appl. Phys. 117, 154306 (2015)

<https://doi.org/10.1063/1.4918316>

CHORUS



Connected Content

This is a companion to: [Cluster dynamics models of irradiation damage accumulation in ferritic iron. I. Trap mediated interstitial cluster diffusion](#)

Articles You May Be Interested In

Cluster dynamics models of irradiation damage accumulation in ferritic iron. I. Trap mediated interstitial cluster diffusion

J. Appl. Phys. (April 2015)

Effect of interatomic potentials on dynamics of supersonic 2-crowdions in tungsten

AIP Conf. Proc. (September 2022)

Radiation-induced phenomena in thermally treated Kr matrices

Low Temp. Phys. (May 2023)

HIDEN
ANALYTICAL

From Gas Phase to Surface:
Diagnostics for Applied Physics Research

Tools for vacuum, plasma processing and experimental physics

Trusted in Research
for over 40 years

Find Solutions for Your Research

Cluster dynamics models of irradiation damage accumulation in ferritic iron. II. Effects of reaction dimensionality

Aaron A. Kohnert and Brian D. Wirth

University of Tennessee, Knoxville, Tennessee 37996-2300, USA

(Received 8 January 2015; accepted 4 April 2015; published online 17 April 2015)

The black dot damage features which develop in iron at low temperatures exhibit significant mobility during *in situ* irradiation experiments via a series of discrete, intermittent, long range hops. By incorporating this mobility into cluster dynamics models, the temperature dependence of such damage structures can be explained with a surprising degree of accuracy. Such motion, however, is one dimensional in nature. This aspect of the physics has not been fully considered in prior models. This article describes one dimensional reaction kinetics in the context of cluster dynamics and applies them to the black dot problem. This allows both a more detailed description of the mechanisms by which defects execute irradiation-induced hops while allowing a full examination of the importance of kinetic assumptions in accurately assessing the development of this irradiation microstructure. Results are presented to demonstrate whether one dimensional diffusion alters the dependence of the defect population on factors such as temperature and defect hop length. Finally, the size of interstitial loops that develop is shown to depend on the extent of the reaction volumes between interstitial clusters, as well as the dimensionality of these interactions. © 2015 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4918316>]

I. INTRODUCTION

Cluster dynamics (CD) models are a powerful reaction rate theory based modeling approach that allow high fidelity calculations of the density of irradiation induced defects and their contributions to long term microstructural evolution.^{1–5} In Paper I,⁵ such models are used to examine the development of so called black dot damage in ferritic alloys, a microstructure that develops at temperatures below approximately 300 °C. The temperature dependence of the black dot density evident from these models excludes any purely thermally activated reaction diffusion mechanism for damage formation. A trap mediated mechanism for the mobility of interstitial type defect clusters proved more successful. This kinetic system envisions crowdion bundles which freely glide along one dimensional trajectories until encountering a trapping site. Such a model is motivated by dynamic observations during ion and electron irradiation reported by *in situ* TEM experiments, during which black dots display significant mobility via a series of discrete hops spanning many nanometers of material.^{6–17}

As typically applied, CD models make a number of assumptions about the interactions between the cluster populations. In the majority of the literature employing such models, all reactions are assumed to be isotropic in nature. This in turn implies that all species in the system diffuse three dimensionally and that the interaction between any given pair of species occurs within a spherically symmetric reaction volume. However, it is well known that the intrinsic migration of crowdion interstitial clusters and prismatic loops occurs one dimensionally and, specifically, in bcc metals along $\langle 111 \rangle$ directions.^{18–22} The apparent importance of a trap mediated kinetics which accounts for this motion suggests that at least the first of these assumptions may be badly

violated. Furthermore, the one dimensional migration of interstitial clusters has long been relied upon to explain a number of aspects of irradiation induced microstructures as an important feature of the production bias model.^{23–26} The assumption of spherical reaction volumes may also become invalid for some reactions, particularly in cases where interstitial clusters grow into loops with diameters exceeding a few nanometers.

In order to more accurately assess the implications of trap mediation, and its ability to explain microstructural development, these assumptions must be relaxed. This study presents a simple framework for incorporating the reactions involving one dimensional diffusers into a CD model. Doing so allows a more explicit representation of the trap mediated interstitial migration, which can only be approximated in prior models. With these improvements, the importance of reaction kinetics in predicting microstructural development can be examined in detail.

II. METHODS

The cluster dynamics model is based on the classical rate theory approach to quantifying the effects of radiation, and a coupled set of reaction-diffusion equations which govern the evolution of each defect cluster population is written as

$$\frac{dC_i}{dt} = \nabla(D_i \nabla C_i) - D_i k_i^2 C_i + R_i(C) + g_i, \quad (1)$$

where C_i is the volumetric concentration of each defect or impurity species i . The terms in these equations represent changes in the density of clusters of a given size as a consequence of spatial diffusion, loss at sinks, interaction with other clusters, and direct production in damage events, respectively.

This approach is applicable regardless of the dimensionality of the diffusers; however, the sink strength k_i^2 and the non-linear reaction term R_i must take different forms for one dimensionally diffusing defects and/or for reactions which do not exhibit spherical symmetry. Regardless of dimensionality, the non-linear loss or gain of a particular defect cluster due to interactions with other clusters $R_i(C)$ is written as a sum over all possible interactions

$$R_i^+(C) = \sum_{n+m \rightarrow i} k_{n,m}^+ C_n C_m - \sum_j k_{i,j}^+ C_i C_j, \quad (2)$$

each with a unique reaction rate coefficient $k_{i,j}^+$. Additionally, clusters may dissociate, releasing one of their members and producing two daughter clusters. The effect on any one cluster population can be written according to

$$R_i^-(C) = \sum_{n \rightarrow n-i,i} k_{n-i,i}^- C_n - \sum_j k_{i,j}^- C_i, \quad (3)$$

where Eqs. (2) and (3) together describe all possible interactions between clusters of different membership. The dissociation constants, k^- , are written as a function of the reaction rate constant for the pair of daughter clusters and the binding energy E^b of the parent

$$k_{i,j}^- = \frac{k_{i-j,j}^+}{\Omega} \exp \frac{-E_{i,j}^b}{k_b T}, \quad (4)$$

where Ω is the atomic volume. Thus, the obstacle to incorporate one dimensional diffusion into a CD model lies in deriving appropriate values of k^+ , which are traditionally derived for three dimensional diffusers. The difference between a reaction rate coefficient and a sink strength is largely a matter of notation. Indeed, the rate $R_{i,j}$ for some set of immobile objects j for absorbing a defect of type i can be written equivalently in terms of the reaction rate coefficient or sink strength as

$$R_{i,j} = D_i k_{i,j}^2 C_i = k_{i,j}^+ C_i C_j, \quad (5)$$

where the second notation simply proves more convenient in describing and implementing the cluster dynamics method.

Existing and well-established descriptions of the sink strength for one dimensionally diffusing defects can be easily translated into reaction rate coefficients. The widely accepted solution

$$k_{i,j}^2 = \frac{2\sigma_{i,j}C_j}{\lambda_i} \quad (6)$$

is derived at length by Borodin,²⁷ where $\sigma_{i,j}$ is the cross section for interaction and λ_i is the mean free path for the one dimensionally diffusing species. The corresponding rate coefficient

$$k_{i,j}^+ = \frac{2D_i\sigma_{i,j}}{\lambda_i} \quad (7)$$

is easily incorporated into a cluster dynamics model. While D_i and $\sigma_{i,j}$ are constant, λ_i will evolve with the defect and sink population, but is easily computed as

$$\lambda_i^{-1} = \Sigma_i = \sum_j \sigma_{i,j} C_j, \quad (8)$$

when written in terms of an effective macroscopic cross section Σ_i . In principle, the rate of reaction is of higher order in the 1d case. In practice, this presents little in the way of additional complexity for numerical integration techniques as Σ_i varies slowly with respect to any particular component of the defect population.

Reactions between a one dimensionally diffusing and three dimensionally diffusing cluster present a more fundamental difficulty. In cases where both defects diffuse three dimensionally, it is legitimate to decompose the full rate coefficient into separate components

$$k_{i,j}^+ = \kappa_{i,j} + \kappa_{j,i}, \quad (9)$$

where $\kappa_{\alpha,\beta}$ is the rate coefficient for α diffusing about stationary β . In this study, we extend this approach to all cases, regardless of dimensionality. This approximation is motivated by an understanding of the conditions of 1d diffusing clusters, and some care should be taken to avoid violating the assumptions which underwrite it. In this application, 1d diffusers are freely gliding crowdion bundles which diffuse much more rapidly than the three dimensional vacancy clusters and trapped prismatic loops they will encounter. The contribution to the reaction rate from three dimensionally diffusing clusters in mixed dimensional situations is negligible. Three dimensionally diffusing clusters which do diffuse at comparable rates will exist in the microstructure at very low concentrations, such that net reaction rate in such cases is itself negligible and the coefficients need not be highly accurate. One dimensional diffusers might interact with one another as well. Following similar logic, the density of trapped loops is far greater than freely diffusing loops, such that the interactions are dominated by the diffusion of free loops and crowdion bundles to trapped ones.

The magnitudes of rate coefficients and sink strengths designed to accommodate one dimensional diffusion are determined primarily by the cross sections. In some instances, characterizing the cross section is straightforward. Consider the limiting case of a one dimensional diffuser approaching a much larger spherical cavity, or void. The cross section can be envisioned as a projection of the reaction volume onto a plane, and is simply a disk with a radius equal to the radius of the reaction volume itself. In cases where the defects have a lower degree of symmetry, the process is more complex. For the point defect—crowdion bundle reaction, the reaction volume resembles a torus for sufficiently large clusters. The projection of this volume results in annular cross sections. This is consistent with the cross sections used by ourselves and others to characterize the trapping interaction for dislocation loops.^{14,15}

The most complex scenario involves reactions between two crowdion bundles. A physically meaningful cross section for interactions of this type is very difficult to determine as the conditions under which two loops will coalesce are governed by complex elastic interactions. Some qualitative insight, however, into the conditions under which crowdion

bundles react has been obtained from molecular dynamics (MD) simulations.^{28,29} A limiting case would be a spherical reaction volume similar to the one used for voids, however it is quite clear that this volume is too large. The strain fields of coaxial interstitial clusters will certainly cause repulsion, for instance, and thus, clusters oriented in this manner should be incapable of coalescence, limiting the cross section. Unless otherwise specified, calculations in this paper will assume annular cross sections, which are formed by the overlap of the cluster dislocation cores. The computation for the cross section is further complicated by the four possible $\langle 111 \rangle$ loop orientations, and the cross section for loops in differing orientations will be compressed significantly. This reduces the magnitude of the cross section, particularly when the diffusing loop is small in comparison to the stationary one.

The formulas used for estimating interaction cross sections in these simulations are outlined in Table I. These involve the reaction of a 1d diffuser with radius r and possible partners including a void of radius r_V , a point defect of radius r_j , and a loop of radius r_L . Additional dimensions include the dislocation capture radius r_d , point defect reaction distance r_0 , trapping radius r_T , and a newly introduced loop coalescence radius r_c . For particularly small values of the diffuser radius, the second term in many of the equations in Table I may become negative, and, in these cases, it is neglected. Otherwise, most of the equations will simplify to $4\pi r r_{(T,d,j,c)}$. In these simulations, r_d and r_c are held equal; however, it is worth noting that they need not be the same in practice. Indeed, the formulas presented here are best interpreted not as a strict representation of a physical reaction volume, but rather as scaling laws between the obstacle size and reaction rate.

The 1d reaction system allows a higher fidelity description of the trap interactions. In the 3d model used in Paper I, the hops induced by irradiation were accounted for by adding an athermal term to the diffusion coefficient. In the 1d context, trapping is implemented as a simple reaction between a freely gliding loop and a trap, treated no differently from any other reaction in the system. The irradiation activated detraping events are implemented as athermal dissociation events with a fixed event frequency ν^{dt} .

For the interaction of three dimensionally migrating species of either vacancy or interstitial type with spherically symmetric clusters, the rate constants take the form

TABLE I. Cross sections for 1d diffusers approaching various types of interaction partners. Aligned loops are defined to share a common Burgers vector. Loops are assumed to be equally distributed between the four possible $\langle 111 \rangle$ orientations and lie on the corresponding $\{111\}$ habit plane.

Interaction type	Cross section
Void	$\pi(r + r_0 + r_V)^2$
Point defect	$\pi(r + r_j + r_d)^2 - \pi(r - r_j - r_d)^2$
Trap	$\pi(r + r_T)^2 - \pi(r - r_T)^2$
Loop (aligned)	$\pi(r + r_L + r_c)^2 - \pi(r + r_L - r_c)^2$
Loop (unaligned)	$\pi(r + r_L + r_c)(r + r_L/3 + r_c)$ $- \pi(r + r_L - r_c)(r + r_L/3 - r_c)$

$k^+ = 4\pi r D$. While this is likely an acceptable representation for vacancy clusters, interstitial clusters develop into loops, implying highly non-spherical reaction volumes, particularly at large loop sizes. The cross sections which were introduced in Table I reflect non-spherical reaction volumes, and an equivalent modification should be made to the rates for three dimensional diffusers as well. A solution for the sink strength of a toroidal reaction volume with a loop radius r_L , interior radius r_p , and density ρ_L , as

$$k_L^2 = \frac{4\pi^2 r_L}{\log(8r_L/r_p)} \rho_L, \quad (10)$$

was developed by Seeger.³⁰ This function, however, is only valid when the loop radius notably exceeds the interior radius as suggested by its creators and confirmed in recent Monte Carlo studies.³¹ Therefore, it is appropriate to use a spherical reaction volume for small loop radii and a toroidal volume for large radii. We transition between the two for increasing loop radius according to

$$k_{ij}^+ = (D_i + D_j)((1 - \alpha_{ij})z_{ij}^L + \alpha_{ij}z_{ij}^V), \quad (11)$$

$$\alpha_{ij} = \left(1 + \left(\frac{r_i}{3(r_j + r_d)}\right)^2\right)^{-1},$$

$$z_{ij}^L = \frac{4\pi^2 r_i}{\log(1 + 8r_i/(r_j + r_d))},$$

$$z_{ij}^V = 4\pi(r_i + r_j + r_d)$$

for a defect j interacting with a loop i . A similar, though not identical, approach has been used by other CD models.⁴ The denominator of Eq. (10) is slightly modified to avoid a singularity at $8r_i = r_j + r_d$, although in cases where this modification is significant the contribution from z^L is screened out. We include this modification for completeness, but its impact on the system is likely to be quite small. It amounts to a rate reduction of less than 20% even at the largest loop sizes considered, whereas 1d kinetics can reduce reaction rates by several orders of magnitude depending on the cross sections relevant to the diffuser in question.

Paper I⁵ considers a variety of irradiation conditions and possible behaviors for small defect clusters. For simplicity, those variables are largely eliminated here to focus on the effects of reaction kinetics alone. Only a heavy ion cascade damage condition is considered, with a primary damage distribution spanning v_9 to i_{20} consistent with that used in Paper I. The material and defect parameters listed in Table II are used unless otherwise specified, and the model permits vacancy clusters with up to 2000 members and interstitial clusters with up to 6000. The table outlines two scenarios for long range vacancy migration, one in which these defects diffuse with the migration energies E_m predicted by *ab initio* calculations³² and one in which the effective vacancy mobility is significantly reduced through the formation of complexes with carbon and other impurity atoms. The same diffusion prefactors D_0 are used in each case. Thin foil and bulk environments are both modeled, the former by applying black absorber

TABLE II. Simulation parameters for cluster dynamics calculations.

Material and irradiation			
Parameter	Value		
Dose rate (dpa/s)	8.6×10^{-4}		
Hop activation, ν^{dt} (dpa $^{-1}$)	30		
Burgers vector, b (nm)	0.249		
Trap density, ρ_T (ppm)	200		
Trap radius, r_T	$1.5b$		
I cluster radius $r_I(n)$	$(\frac{\Omega n}{\pi b})^{1/2}$		
V cluster radius $r_V(n)$	$(\frac{3\Omega n}{4\pi})^{1/3}$		
Dislocation bias β	1.1		
Dislocation density ρ_d (nm $^{-2}$)	2.5×10^{-4}		
Foil thickness (nm)	100		
Diffusion			
Cluster	prefactor (nm 2 /s)	E _m (eV) Set I	Set II
v_5+	0	...	...
v_4	$D_{0,v1}/4$	1.3	0.48 (Ref. 32)
v_3	$D_{0,v1}/3$	1.3	0.35 (Ref. 32)
v_2	$D_{0,v1}/2$	1.3	0.62 (Ref. 32)
v_1	10^{11}	1.3	0.67 (Ref. 32)
i_1	10^{11}	0.34	0.34 (Ref. 33)
i_2	$D_{0,i1}/2$	0.42	0.42 (Ref. 33)
i_3	$D_{0,i1}/3$	0.43	0.43 (Ref. 32)
$i_4 - i_\infty$ (trapped)	$D_{0,i1}/n^{0.7}$	1.3	1.3
$i_4 - i_\infty$ (free)	$D_{0,i1}/n^{0.7}$	0.1	0.1

boundary conditions to a spatial discretization of Eq. (1) and the latter by eliminating spatial dependence entirely.

The one dimensional system allows a more detailed examination of the thermal mobility of crowdion bundles and prismatic loops. In the three dimensional model, the thermal activation energy was implemented as a migration energy. This interpretation can be extended to the new model by implementing the activation energy as a migration energy for the trapped clusters. Alternatively, it can be applied as a dissociation energy between a freely gliding loop and a trap. Using this approach, the thermally activated migration process is trap mediated in the same manner as the beam activated hops. Both approaches will be examined, though the latter is used by default. In either case, trap interaction is permitted for all 1d diffusers. In all cases, interstitial clusters with four or more members are considered to diffuse purely one dimensionally, while smaller interstitial clusters and vacancy clusters diffuse in three dimensions.

III. RESULTS

A. Visible defect density

The introduction of interstitial cluster mobility through discrete hops between traps had a number of implications on the evolution of defect clusters in a three dimensional diffuser model. Chief among these was an insensitivity to temperature below roughly 300 °C with a loop density that was largely independent of vacancy mobility. The temperature dependence of the trap mediated diffusion system was not strongly

impacted by a change in the dimensionality of reaction kinetics. The peak loop density as a function of temperature with 1d kinetics and migration set I is shown in Figure 1. When the trapped loops were given a thermal mobility identical to that of the isotropic model as specified in Table II, the transition to thermally dependent regime was largely similar to that seen with a 3d model. When the model considered the thermal motion of loops by allowing thermal dissociation from traps instead of the migration of trapped loops themselves, the athermal region terminated at lower temperatures. Interestingly, the prefactor reported by Arakawa for loop diffusion is roughly four orders of magnitude higher than that for crowdion bundles.³⁴ When an equivalent modification was made to the prefactor for trapped loops in the thermal migration model, the same temperature shift occurred.

Figure 1 also demonstrates the impact of vacancy mobility on the loop density. When *ab initio* values were used for long range vacancy migration, the peak loop density reached in the model dropped by nearly an order of magnitude between 0 and 100 °C. This represents the most significant deviation from the behavior of the 3d model in terms of temperature dependence. In that model, different assumptions about the vacancy mobility had little impact on the defect density, and primarily affected the growth rate of visible defect clusters. The transition to a strong thermal dependence is unaffected by vacancy migration, and occurs near 200 °C when the thermal activation of loops is assumed to be the result of dissociation from traps.

Using set I, the defect evolution below 100 °C was largely independent of temperature in the trap mediation model regardless of assumptions about the dimensionality of diffusion or the mechanism by which loops become thermally mobile. Though nucleation behavior is independent of these

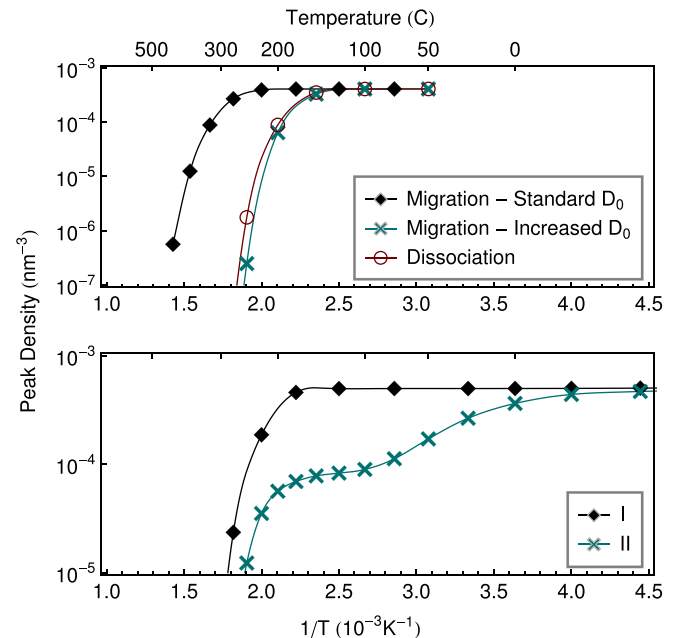


FIG. 1. Saturation density of interstitial clusters larger than 2.5 nm in diameter as a function of temperature assuming 1d reaction kinetics. The thermal mobility of trapped loops was tested for two different prefactors and compared to dissociation from traps (top), and the effect of vacancy migration was assessed using both migration energy sets from Table II (bottom).

factors, the long term evolution differs substantially between assumptions about reaction dimensionality. We have thoroughly investigated the effect that one dimensional kinetics has on defect evolution at room temperature. One example of the differences between the two systems is shown in Figure 2, which examines the relationship between trap and defect densities. The 1d system displayed significantly lower sensitivity to the trap density than the 3d system in all cases.

In a thin foil environment, the dependence on trap density was of approximately orders 1 and $1/2$ at high densities for 3d and 1d kinetics, respectively. With the decrease in loop density, the scaling order slowly increased, reaching upper thresholds at twice the high density limit. The same calculations were repeated in a bulk environment with no dislocation sinks. A deviation from the high density trend was not observed in this case, and a constant power law behavior was observed for the entire range examined. One notable exception occurred in the 1d kinetics calculations, where the loop density ultimately became limited by the number of traps itself at sufficiently low trap density.

In all cases, a maximum value of the loop density was reached prior to 1 dpa. The dose at which saturation occurred was delayed by an increased trap density, and occurred later for 1d kinetics than 3d kinetics. Following saturation, the number density slowly declined regardless of reaction dimensionality. This occurred even in simulations without the presence of surfaces or external sinks, indicating that interaction with other defects plays a strong role in limiting the visible defect cluster population.

The loop depletion caused by sinks and surfaces can also be examined in terms of foils of different thickness. Figure 3 shows the maximum depth averaged loop concentration in foils of various thickness as compared to the concentration that develops in a bulk environment. Three values of trap density were examined, and the loop concentration is substantially reduced in sufficiently thin foils. The minimum thickness at which loops appear in notable concentrations

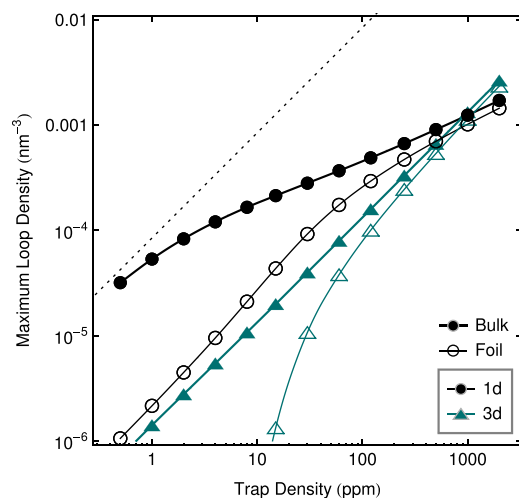


FIG. 2. Effect of trap density on maximum loop concentrations for three dimensional and one dimensional reaction kinetics assumptions at 300 K. Open points correspond with the standard thin foil environment while filled point denote a system with no surfaces or dislocation sinks. The dotted line indicates a loop density equivalent to the trap density.

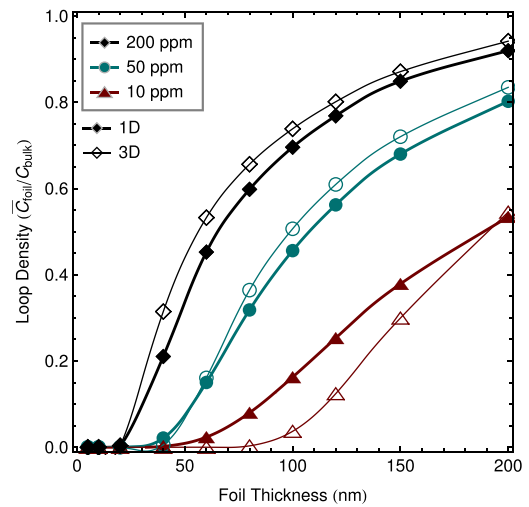


FIG. 3. Maximum density of observable loops as a function of foil thickness relative to the density in a bulk calculation at 300 K. Three trap densities are considered with both 1d (filled points) and 3d kinetics (open points).

varied by trap density. This effect occurred with both sets of reaction kinetics with only subtle differences. In each case, three dimensional kinetics produced a sharper dependence on thickness and loops first appeared in concentrations comparable to the bulk at thicker foil sizes. These differences became more pronounced at lower trap densities.

B. Defect growth and size distributions

Reaction dimensionality as well as the shape and size of the reaction volumes had a strong impact on the long term evolution of the defect population. Figures 4 and 5 illustrate the extent to which reaction kinetics influence the nature of the defects that arise from trap mediated interstitial motion. Three distinct set of assumptions regarding the reaction kinetics were considered. The first was a three dimensional system with spherical reaction volumes equivalent to that examined in our previous study.⁵ The second considered one dimensional kinetics, but with cross sections representing spherical interactions. The final system also examined one dimensional kinetics, but using the non-spherical, dislocation based cross section parametrization presented in Table I. In all cases, the characteristic reaction distances r_0 , r_d , and r_c were all equal to a single parameter, denoted here as the critical distance.

The 1d systems produced higher densities than the 3d system at a trap density of 200 ppm. The magnitude of this difference will scale with trap density as demonstrated earlier. All three systems demonstrated a decrease in saturation density as the size of the reaction volume was increased, and each declined by a factor of nearly 2.5 between the smallest and largest reaction volumes examined. The only exception to this appeared at very small critical distances in the dislocation system, where an appreciable loop population failed to emerge.

In contrast to density, the loop growth characteristics depended strongly on the kinetic assumptions. At the smallest considered reaction volumes, the 1d system with spherical cross sections produced loops approaching 16 nm within

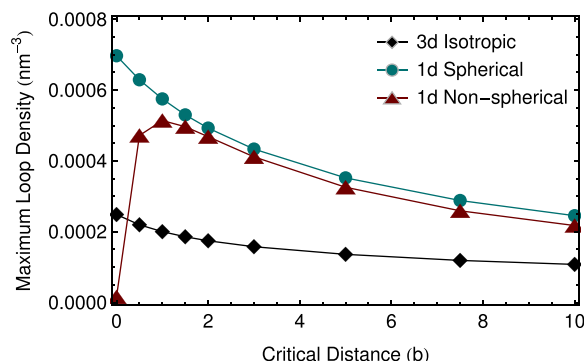


FIG. 4. Calculated saturation density at 300 K for ion irradiation with differing reaction kinetics assumptions.

the first dpa whereas the 3d system produced slightly smaller loops. By comparison, the dislocation cross sections resulted in loops barely above the visible size threshold. The terminal loop size of loops in the dislocation system increased with the reaction distances only at small values, and the growth rate was largely independent of these parameters at larger values. The 3d system produced larger loops, and the opposite trend with respect to interaction distances was observed. Terminal loop size in the spherical cross section system also declined with increasing interaction distances. The magnitude of this dependence was larger than for the other systems, and showed no sign of reaching a stable value.

A restriction on the number of equations permissible in this model restricted the maximum cluster size to 6000 members, equating to a loop roughly 19 nm in diameter. While significant concentrations of loops were not usually developed at this boundary by 1 dpa, the case of 1d diffusion with loops as spherical absorbers presented an exception. This is illustrated by the size distributions of loops shown in Figure 6. Because larger clusters are unavailable to the model, the mean cluster size is artificially constrained in simulations where spherical cross sections are used. In this manner, the large loop sizes and strong dependence on reaction parameters that characterize this system are understated in Figure 5.

The loop size distributions also reveal a fundamental difference in the way in which the defect population evolves

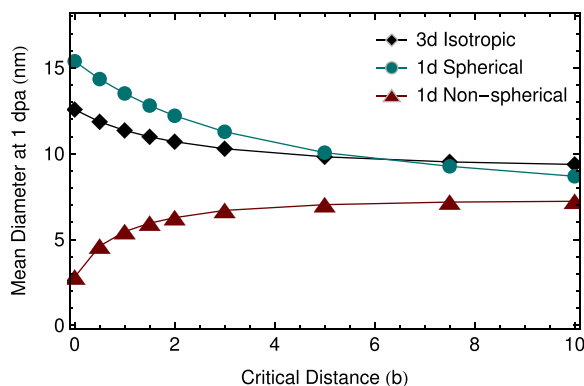


FIG. 5. Calculated loop sizes following 1 dpa of ion irradiation at 300 K. The plotted values represent the mean diameter of all loops larger than 2.5 nm.

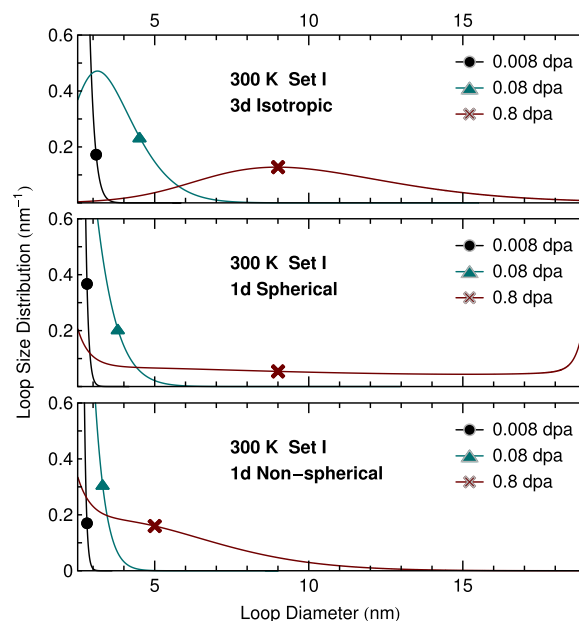


FIG. 6. Size distributions of loops larger than 2.5 nm demonstrating the evolution with dose predicted by various reaction kinetic assumptions.

when the reaction dimensionality changes. With three dimensional mobility for loops, the size distribution is peaked near the mean diameter and takes a roughly normal shape. The distribution peak shifts to larger sizes and broadens with increasing dose, consistent with the familiar nucleation and growth scenario. The behavior with 1d kinetics is quite different. Using either set of cross sections, the greatest number density of defects is found at small sizes, even as the average size continues to increase. Though the rate of this increase is strongly dependent on the cross section for absorption, the dense population of nuclei is never depleted. The loops which grow to large sizes appear as a long tail on the size distribution and constitute only a fraction of the total population.

The loop growth and size distributions produced by set I were largely independent of temperature, up to the point where loop motion becomes thermally activated. This was not the case in set II, where the effects of vacancy mobility become evident near room temperature. Growth was noticeably slowed by increasing temperature, and the size distributions became very strongly peaked near the visibility threshold throughout the duration of the simulation as shown in Figure 7. In addition, the tail of the size distribution toward larger cluster sizes was shortened with increasing temperature. In some conditions, the average size of visible clusters had begun to decline slightly within 1 dpa of simulation. None of these effects were observed in set I.

The degree to which vacancy mobility suppressed the formation of larger defects depended strongly on the interaction distances. At larger reaction volume sizes, the tail of the size distribution to large sizes still developed. In these cases, the form of the size distributions produced by sets I and II differed only slightly. In both cases, the size distributions that developed became insensitive to the interaction distances past a few nearest neighbor spacings, consistent with the behavior of the loop terminal size.

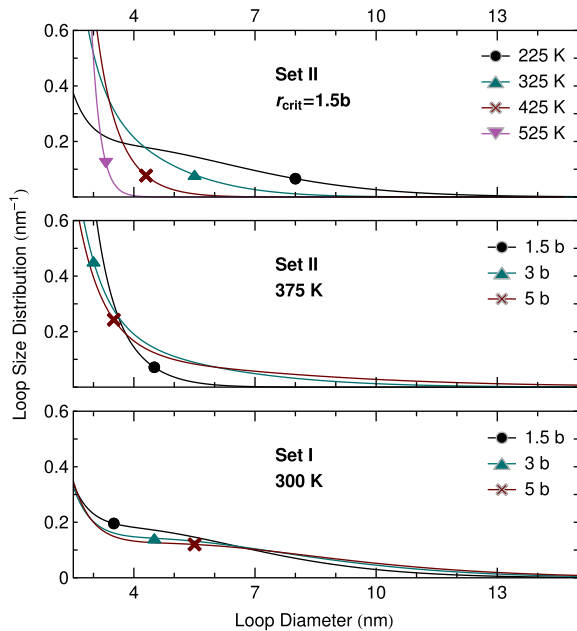


FIG. 7. Impact of vacancy mobility on size distributions of loops larger than 2.5 nm when non-spherical reaction volumes are considered. The effects of temperature (top) and reaction distance (middle) are shown along with size distributions where vacancies are effectively immobile (bottom).

IV. DISCUSSION

A. Effects of 1d kinetics

In Paper I, we demonstrated that the transition between thermal and athermal behavior in loop density in our model mirrored experimental trends under thin-film ion irradiation in iron-chrome based alloys. While 1d kinetics alone do not significantly alter this picture, uncertainty about the means of thermal mobility for defects produce a range of possible temperatures for this transition to occur. A mechanism of simple thermal migration by site to site jumps for trapped loops as we previously assumed produces a transition temperature nearer to that seen in the iron chrome alloys, but thermal dissociation from traps produces something more in line with the prefactor suggested by quantitative measurements of high temperature loop motion. These measurements were made in high purity iron, however, and the presence of chrome may further hinder loop motion and possibly raise the effective activation energy for defect mobility. Further work is needed to analyze the difference between defect evolution in high purity iron and iron chrome alloys along these lines. We also note that the temperature at which this transition occurs will be dose rate sensitive, with thermal effects taking hold at lower temperatures if the dose rates were decreased.

The remainder of this discussion is confined to the behavior at lower temperatures, where traps and irradiation induced detrapping drive changes in the defect population in our model. The scaling that emerges with respect to trap density is an entirely expected consequence of the reaction kinetics between potential loop nuclei. At an equilibrium concentration of nuclei, the rate at which they are generated in cascades will be equal to the rate at which they agglomerate through mutual interaction, such that in general terms

$g \approx k^+ C^2$. With 1d and 3d reaction kinetics, $1/k^+$ is proportional to the trap density and the trap density squared, respectively, giving rise to the observed scaling. When the fate of cascade-born clusters is dominated by absorption on surfaces and sinks rather than agglomeration with other defects, the equilibrium condition changes to $g \approx Dk^2 C$, and the order at which nuclei concentrations scale with traps is higher.

The weaker dependence on trap density which emerges from 1d kinetics has interesting consequences for the expected behavior of ion irradiated bcc iron alloys. Namely, the defect concentration which develops should not be strongly dependent on the alloy examined if irradiation induced detrapping is indeed the driving force for evolution. The ultimate density of black dots would not be expected to vary by more than an order of magnitude between high purity model iron and commercial alloys. According to this model, detrapping dominates the evolution kinetics near room temperature, and the microstructure that develops would be independent of dose rate. Indeed, the defect population should be more prominently influenced by the damage source, as demonstrated in our previous comparison of heavy ion and electron irradiation. The lower energy recoils associated with proton irradiation for instance, will generate fewer large interstitial clusters directly in cascades and might also alter the detrapping rate when compared to heavy ion irradiation.

Another interesting consequence of trap mediation is the depletion of interstitial loops in, particularly, thin regions of the foil. This would not be expected in a scenario where interstitial loop nuclei were entirely immobile. Any defects which appear in thin regions of the foil, particularly in high purity samples, must not be $\langle 111 \rangle$ interstitial type loops according to this model. A possible exception exists in $\langle 110 \rangle$ oriented foils, where half of the possible glide cylinders never intersect the surface. Even in this case, the easy extinction of other crowdion clusters at the surfaces will leave a vacancy surplus which would at least slow, if not forbid, the agglomeration of interstitials into visibly sized loops. Any visible defects which develop in thinner foils would be expected to be of vacancy type, or possibly $\langle 100 \rangle$ interstitial loops.

Visible loops which develop in thicker foils show a stronger dependence on how the reaction cross sections scale with loop size than the parameters which define the scaling laws. Clearly, the rate of growth and size distributions of loops varied dramatically when spherical reaction geometries were assumed rather than annular ones, while changing the interaction distances associated with each had much less pronounced effects. This can be explained as a consequence of how cascade generated crowdion bundles are partitioned between existing clusters. The reaction distance impacts the cross section for all interactions equally, or at least nearly so, but assumptions about the reaction volume geometry effect the ratios between cross sections for defects of different sizes. One notable exception occurs when the critical distances for cluster interaction are assumed to be zero, the loop growth rate becomes sufficiently slow that no loops reach visible sizes at all. The cross sections for interaction between crowdion bundles and for loops absorbing crowdion bundles

vanish in this exceptional case. This reveals that the appearance of visibly sized interstitial loops crucially hinges on the ability of loop nuclei to absorb crowdion bundles and possibly coalesce with one another.

The widely differing size distributions produced by 1d and 3d kinetics can be understood in terms of a change in the lifetimes and partitioning of crowdion bundles. The rate of loop growth depends on whether interstitial clusters are more likely to interact with other features in the microstructure as opposed to one another. The nature of 1d reaction kinetics causes a positive feedback cycle in this regard. As the loops grow, their absorption cross sections grow with them, and the cascade generated crowdion bundles are increasingly partitioned to the largest loops which further accelerates loop growth. Growth rate becomes a strong function of size, promoting the largest clusters to ever larger sizes while smaller, barely visible defects grow slowly, if at all. This is particularly evident in the size distribution when spherical reaction volumes are considered for loops. The effect is less pronounced with dislocation based cross sections, but such behavior still informs the shape of the size distributions that emerge. Furthermore, the lifetimes of these small clusters and their sub-visible counterparts are considerably longer under 1d kinetics as a consequence of small reaction cross sections and long mean free paths, which imply slower rates of reaction. The low interaction rates which arise from 1d kinetics allow small loops to persist in the microstructure, even when mobile. Qualitatively, the size distributions produced in this model reflect a high density of small (2 to 4 nm) loops co-existent with a smaller population of growing defects.

B. Comparison to experiment

A number of ion and neutron studies at these temperatures indicate size distributions peaked at or very near the smallest observable sizes.^{9,10,35} Despite this, we have avoided direct quantitative comparisons of the 1d model to experimental data at this juncture because the visibility criteria become particularly consequential. In our data, a sharp cutoff at 2.5 nm delineates visible from invisible defect clusters, but the size profiles indicate that a large fraction of the defect clusters are below this threshold throughout the dose increments simulated. Thus, we expect experimentally reported data on both the cluster density and mean diameter to be strongly sensitive to how many of the small defects are detected and the amount of care taken in measuring their size. Adjusting the visibility threshold to include smaller defect sizes causes the model to predict earlier saturation times while displaying a higher density of defects with reduced mean size.

Such considerations aside, resolvable loops exceeding 10 nm in diameter do not typically develop during *in-situ* experiments at room temperature and below. Rather, damage remains in the form of black dots with sizes nearer to the visibility limit. Calculations based on the trap mediation model as presented here do not reflect such behavior. Indeed, the steady loop growth evident in 3d trap mediation model motivated the introduction of 1d kinetics. The size distributions that emerge from the 1d model remain peaked at small sizes throughout the simulation, and this demonstrates significant

improvement with experiment over the fully 3d model. This apparently cannot resolve all discrepancies, as some larger features develop and grow. A number of possibilities might reconcile the remaining inconsistencies.

First, we examine the ability of vacancy mobility to suppress the development of larger loops. Mobility of vacancies also has the effect of reducing the peak defect density that develops, which is typically higher in magnitude in these models than what is experimentally observed. Were mobile vacancies indeed solely responsible for suppressing loop growth however, the model predicts that large loops should emerge below temperatures where vacancy migration begins. Such an effect is not observed. In fact, to the extent that any temperature dependence in defect size can be deduced from *in situ* ion irradiation, it favors the opposite trend. Additionally, the effectiveness of this mechanism is sensitive to the size of the reaction cross sections, and it could only operate if interaction distances are within roughly the fourth nearest neighbor. Due to the parameter sensitivity of this effect and its inability to operate at low temperature, we consider it an unreliable explanation of the experimentally observed size profiles.

Alternatively, when vacancies are immobile, they accumulate to rather high densities in the microstructure. Irradiation induced motions in the 3d model result in frequent reactions with immobile defects and limit the net vacancy inventory to a few hundred ppm, including those in clusters. With such a vacancy density, it might be tempting to assume that the vacancy sink strength should be strong enough to preclude any evolution of the loop population altogether. This clearly does not occur, primarily because the sink strength for loops is also quite high for the defect densities typical here. Taking a lower bound of the loop density as 10^{-5} nm^{-3} , order of magnitude estimates of sink strength take 10^{17} m^{-2} for vacancies and at least 10^{14} m^{-2} for loops. This leaves loops free to grow by absorption of one of every thousand produced interstitials, hence the growth evident in our models. Note that these models assume that all the recombination and clustering processes are diffusion limited, which may not be the case when the defect density becomes this dense. For instance, the existing defects may reduce the damage efficiency of cascades to the extent that further evolution of the microstructure slows. Unfortunately, investigating such a possibility requires a complex, interdependent coupling between the disparate scales of long term evolution and picosecond damage production, and as such is extraordinarily difficult in practice. Finally, we note that although vacancy mobility and vacancy accumulation are mutually exclusive, it is conceivable that each operates in its own temperature regime.

The final possibility is that the large growing features evident in these models do not correspond to individual loops, but rather to rafts which form through the interaction of multiple component clusters. Examining this possibility quantitatively would require a more sophisticated model. However, the evolution of the size distributions produced by the 1d model is qualitatively consistent with the process of raft formation. A high number density of defects remain at small sizes throughout the simulation while the density of growing defect structures is much smaller. The continuing

growth process of the larger features progresses as initially dense population of smaller clusters is slowly depleted. Additional modeling work is required, treating the interactions of visibly sized loops explicitly as a rafting rather than coalescence event to assess the viability of this interpretation.

V. CONCLUSIONS

One dimensional reaction kinetics have been incorporated into the trap mediated cluster dynamics model of damage evolution, allowing a more natural description of trapping interactions, and a more physically accurate assessment of interstitial cluster interactions in general. The one dimensional nature of cluster motion in and of itself has little impact on the temperature dependence of loop formation, and trap mediation with beam assisted detrapping remains essential to capture the observed temperature dependence of black dot density in thin-film ion irradiation studies. The mechanism by which visible defects achieve thermal mobility can affect the defect population, however, and this can only be examined in the context of 1d reaction kinetics. The dimensionality of reaction kinetics strongly impacts the size and size distribution of the defect clusters which emerge in the low temperature regime. Three dimensional kinetics produces a roughly normally distributed size distribution which grows steadily with increasing dose. One dimensional kinetics, on the other hand, give rise to a large population of small defects with sizes near the visibility threshold alongside a population of growing clusters. The growth of these features at low temperatures is subject to assumptions regarding the interaction cross sections between interstitial clusters, and will require further coordinated modeling and experimental studies to fully resolve.

ACKNOWLEDGMENTS

The authors gratefully acknowledge financial support from the U.S. Department of Energy, Office of Fusion Energy Sciences under Grant No. DOE-DE-SC0006661 and the U.S. Department of Energy, Office of Nuclear Energy, Nuclear Energy University Programs (NEUP).

¹M. Mathon, A. Barbu, F. Dunstetter, F. Maury, N. Lorenzelli, and C. de Novion, *J. Nucl. Mater.* **245**, 224 (1997).

- ²D. Xu, B. D. Wirth, M. Li, and M. A. Kirk, *Acta Mater.* **60**, 4286 (2012).
- ³E. Meslin, A. Barbu, L. Boulanger, B. Radiguet, P. Pareige, K. Arakawa, and C. Fu, *J. Nucl. Mater.* **382**, 190 (2008).
- ⁴A. H. Duparc, C. Moingeon, N. S. de Grande, and A. Barbu, *J. Nucl. Mater.* **302**, 143 (2002).
- ⁵A. Kohnert and B. Wirth, *J. Appl. Phys.* **117**, 154305 (2015).
- ⁶M. Hernandez-Mayoral, Z. Yao, M. L. Jenkins, and M. A. Kirk, *Philos. Mag.* **88**, 2881 (2008).
- ⁷Z. Yao, M. Hernandez-Mayoral, M. Jenkins, and M. Kirk, *Philos. Mag.* **88**, 2851 (2008).
- ⁸Z. Yao, M. Jenkins, M. Hernández-Mayoral, and M. Kirk, *Philos. Mag.* **90**, 4623 (2010).
- ⁹C. Topbasi, A. T. Motta, and M. A. Kirk, *J. Nucl. Mater.* **425**, 48 (2012).
- ¹⁰D. Kaoumi, J. Adamson, and M. Kirk, *J. Nucl. Mater.* **445**, 12 (2014).
- ¹¹M. A. Kirk, P. M. Baldo, A. C. Liu, E. A. Ryan, R. C. Birtcher, Z. Yao, S. Xu, M. L. Jenkins, M. Hernandez-Mayoral, D. Kaoumi, and A. T. Motta, *Microsc. Res. Tech.* **72**, 182 (2009).
- ¹²M. Jenkins, Z. Yao, M. Hernández-Mayoral, and M. Kirk, *J. Nucl. Mater.* **389**, 197 (2009).
- ¹³K. Arakawa, M. Hatanaka, H. Mori, and K. Ono, *J. Nucl. Mater.* **329–333**, 1194 (2004).
- ¹⁴Y. Satoh, H. Matsui, and T. Hamaoka, *Phys. Rev. B* **77**, 094135 (2008).
- ¹⁵Y. Satoh and H. Matsui, *Philos. Mag.* **89**, 1489 (2009).
- ¹⁶T. Hamaoka, Y. Satoh, and H. Matsui, *J. Nucl. Mater.* **433**, 180 (2013).
- ¹⁷K. Arakawa, H. Mori, and K. Ono, *J. Nucl. Mater.* **307–311**, 272 (2002).
- ¹⁸B. Wirth, G. Odette, D. Maroudas, and G. Lucas, *J. Nucl. Mater.* **244**, 185 (1997).
- ¹⁹B. Wirth, G. Odette, D. Maroudas, and G. Lucas, *J. Nucl. Mater.* **276**, 33 (2000).
- ²⁰Y. Osetsky, D. Bacon, A. Serra, B. Singh, and S. Golubov, *J. Nucl. Mater.* **276**, 65 (2000).
- ²¹Y. N. Osetsky, D. J. Bacon, A. Serra, B. N. Singh, and S. I. Golubov, *Philos. Mag.* **83**, 61 (2003).
- ²²N. Soneida and T. D. de La Rubia, *Philos. Mag. A* **81**, 331 (2001).
- ²³H. Trinkaus, B. Singh, and A. Foreman, *J. Nucl. Mater.* **199**, 1 (1992).
- ²⁴B. Singh, S. Golubov, H. Trinkaus, A. Serra, Y. Osetsky, and A. Barashev, *J. Nucl. Mater.* **251**, 107 (1997).
- ²⁵H. Trinkaus, B. Singh, and S. Golubov, *J. Nucl. Mater.* **283–287**, 89 (2000).
- ²⁶A. Barashev and S. Golubov, *Philos. Mag.* **89**, 2833 (2009).
- ²⁷V. Borodin, *Physica A* **260**, 467 (1998).
- ²⁸Y. Osetsky, A. Serra, and V. Priego, *J. Nucl. Mater.* **276**, 202 (2000).
- ²⁹Y. Osetsky, D. Bacon, B. Singh, and B. Wirth, *J. Nucl. Mater.* **307–311**, 852 (2002).
- ³⁰A. Seeger and U. Gosele, *Phys. Lett. A* **61**, 423 (1977).
- ³¹V. Jansson, L. Malerba, A. D. Backer, C. Becquart, and C. Domain, *J. Nucl. Mater.* **442**, 218 (2013).
- ³²C. C. Fu, J. D. Torre, F. Willaime, J. L. Bocquet, and A. Barbu, *Nat. Mater.* **4**, 68 (2005).
- ³³C. C. Fu, F. Willaime, and P. Ordejon, *Phys. Rev. Lett.* **92**, 175503 (2004).
- ³⁴K. Arakawa, K. Ono, M. Isshiki, K. Mimura, M. Uchikoshi, and H. Mori, *Science* **318**, 956 (2007).
- ³⁵M. Victoria, N. Baluc, C. Bailat, Y. Dai, M. Luppó, R. Schaublin, and B. Singh, *J. Nucl. Mater.* **276**, 114 (2000).